2026-2027 University Catalog

CATALOG SEARCH

Entire Catalog

Select Location(s)

Search Catalog

Whole Word/Phrase

Advanced Search

Catalog Navigation

Aeronautical and Astronautical Engineering, BSAAE

Location(s): West Lafayette

Return to: Catalog Search

About the Program

The BS program in Aeronautical and Astronautical Engineering is accredited by the Engineering Accreditation Commission of ABET, under the commission's General Criteria and Program Criteria for Aerospace and Similarly Named Engineering Programs. The General and Program Criteria prescribed by the Engineering Accreditation Commission can be viewed ABET..

School of Aeronautics and Astronautics

Aeronautical and Astronautical Engineering Major Change (CODO) Requirements

Degree Requirements

130 Credits Required

Aeronautical and Astronautical Major Requirements (62 credits)

A GPA of 2.0 is required for all courses in Major Requirements section. This includes Aeronautical and Astronautical Engineering Major, Specialization, and Selective courses.

Aeronautical and Astronautical Engineering Major Courses (41 credits)

- AAE 20000 - Undergraduate Sophomore Seminar **Credit Hours: 0.00**
- AAE 20300 - Aeromechanics I **Credit Hours: 3.00** ♦ (C- or better)
- AAE 20400 - Aeromechanics II **Credit Hours: 3.00** ♦ (C- or better)
- AAE 20401 - Aeromechanics II Laboratory **Credit Hours: 1.00**
- AAE 25100 - Introduction To Aerospace Design **Credit Hours: 3.00** ♦
- AAE 30000 - Undergraduate Junior Seminar **Credit Hours: 0.00**
- AAE 30100 - Signal Analysis For Aerospace Engineering **Credit Hours: 3.00**
- AAE 33300 - Fluid Mechanics **Credit Hours: 3.00**
- AAE 33301 - Fluid Mechanics Laboratory **Credit Hours: 1.00**
- AAE 33400 - Aerodynamics **Credit Hours: 3.00**
- AAE 34000 - Dynamics And Vibrations **Credit Hours: 3.00**
- AAE 35200 - Structural Analysis I **Credit Hours: 3.00**
- AAE 36400 - Control System Analysis **Credit Hours: 3.00**
- AAE 36401 - Control Systems Laboratory **Credit Hours: 1.00**
- AAE 40000 - Undergraduate Senior Seminar **Credit Hours: 1.00**

- AAE 33401 - Aerodynamics Laboratory **Credit Hours: 1.00** or
- AAE 35201 - Structural Analysis I Laboratory **Credit Hours: 1.00**

- AAE 33800 - Thermal Sciences **Credit Hours: 3.00** (C- or better) or

- AAE 33900 - Aerospace Propulsion **Credit Hours: 3.00**

- AAE 42100 - Flight Dynamics And Control **Credit Hours: 3.00** or
- AAE 44000 - Spacecraft Attitude Dynamics **Credit Hours: 3.00**

- AAE 45000 - Spacecraft Design **Credit Hours: 3.00** or
- AAE 45100 - Aircraft Design **Credit Hours: 3.00**

Aeronautical and Astronautical Engineering Selectives (21 credits)

See Supplemental Information for approved courses.

- AAE Engr Specialization - Credit Hours: 9.00
- AAE Selectives - Credit Hours: 6.00
- AAE Technical Selective - Credit Hours: 3.00
- AAE Statistics Selective - Credit Hours: 3.00

Other Departmental/Program Course Requirements (68-73 credits)

Other Course Requirements (44-48 credits)

Intro to Engineering I & II - Credit Hours: 4.00-8.00

- ENGR 13100 - Transforming Ideas To Innovation I (UCC: AI/IL) **Credit Hours: 2.00** and
- ENGR 13200 - Transforming Ideas To Innovation II (UCC: AI) **Credit Hours: 2.00**

  OR
- EPCS 11100 - First Year Participation In EPICS I (UCC: STS) **Credit Hours: 1.00** and
- EPCS 12100 - First Year Participation In EPICS II (UCC: STS) **Credit Hours: 1.00** and
- ENGR 13300 - Transforming Ideas To Innovation, EPICS/VIP (UCC: AI/IL) **Credit Hours: 2.00**

  OR
- VIP 17911 - First Year Participation In Vertically Integrated Projects (VIP) I (UCC: STS) **Credit Hours: 1.00** and
- VIP 17912 - First Year Participation In Vertically Integrated Projects (VIP) II **Credit Hours: 1.00** and

- ENGR 13300 - Transforming Ideas To Innovation, EPICS/VIP (UCC: AI/IL) - 2.00 **Credits Hours**

  OR
- ENGR 16100 - Honors Introduction To Innovation And The Physical Science Of Engineering Design I (UCC: AI/IL) **Credit Hours: 4.00** and
- ENGR 16200 - Honors Introduction To Innovation And The Physical Science Of Engineering Design II (UCC: SCI) **Credit Hours: 4.00**

  OR
- ENGR 13000 - Transforming Ideas Into Innovations (UCC: AI/IL) **Credit Hours: 4.00**

- CHM 11510 - General Chemistry I **Credit Hours: 3.00**

  AND
- CHM 11520 - General Chemistry I - Laboratory (UCC: SCI) **Credit Hours: 1.00** or
- CHM 11530 - General Chemistry I - Virtual Laboratory (UCC: SCI) **Credit Hours: 1.00**

- MA 16100 - Plane Analytic Geometry And Calculus I (UCC: QR) **Credit Hours: 5.00** or
- MA 16500 - Analytic Geometry And Calculus I (UCC: QR) **Credit Hours: 4.00**

- MA 16200 - Plane Analytic Geometry And Calculus II (UCC: QR) **Credit Hours: 5.00** or
- MA 16600 - Analytic Geometry And Calculus II (UCC: QR) **Credit Hours: 4.00**

Physics I - Credit Hours:0.00-4.00

- PHYS 17200 - Modern Mechanics (UCC: SCI) **Credit Hours: 4.00**

  OR
- ENGR 16100 - Honors Introduction To Innovation And The Physical Science Of Engineering Design I (UCC: AI/IL) - 4.00 **Credit Hours** and
  - ENGR 16200 - Honors Introduction To Innovation And The Physical Science Of Engineering Design II (UCC: SCI) - 4.00 **Credit Hours**

CS Requirement - Credit Hours: 3.00-4.00

- CS 15900 - C Programming **Credit Hours: 3.00**
- CS 17700 - Programming With Multimedia Objects **Credit Hours: 4.00**
- CS 18000 - Problem Solving And Object-Oriented Programming **Credit Hours: 4.00**

- MA 26100 - Multivariate Calculus (UCC: QR) **Credit Hours: 4.00** ♦
- MA 26500 - Linear Algebra (UCC: QR) **Credit Hours: 3.00** ♦
- MA 26600 - Ordinary Differential Equations (UCC: QR) **Credit Hours: 3.00** ♦

- MA 30300 - Differential Equations And Partial Differential Equations For Engineering And The Sciences **Credit Hours: 3.00** ♦
- ME 20000 - Thermodynamics I **Credit Hours: 3.00** ♦
- MFET 16300 - Foundations In Geometric Modeling And Data Management **Credit Hours: 2.00**

- PHYS 24100 - Electricity And Optics (UCC: SCI) **Credit Hours: 3.00** or
- PHYS 27200 - Electric And Magnetic Interactions (UCC: SCI) **Credit Hours: 4.00**

Engineering General Education Requirements (24-25 credits)

For a list of courses that satisfy this requirement, see Supplemental Information

- General Education I (UCC: HUM) - Credit Hours: 3.00-4.00
- General Education II (UCC: BSS) - Credit Hours: 3.00
- General Education III - Credit Hours: 3.00
- General Education IV - Credit Hours: 3.00
- General Education V - Credit Hours: 3.00
- General Education VI - Credit Hours: 3.00
- Written Communication (UCC: WC) - Credit Hours: 3.00-4.00
- Oral Communication (UCC: OC) - Credit Hours: 3.00

Additional General Education Requirements

- At least 6 credits must be in non-introductory courses (30000-level or above OR from courses with a required pre-requisite in the same department)
- At least 12 credits must be taken from the College of Liberal Arts, the Daniels School of Business, and/or the Honors College provided such courses are not focused primarily on engineering, technology, the natural sciences, or mathematics.
- A course is not eligible for use towards these 24 credits of Engineering General Education requirement if that course is used to satisfy a different engineering degree requirement.

Supplemental Information

Click here for Aeronautical and Astronautical Engineering Supplemental Information sheet.

GPA Requirements

- A cumulative GPA of 2.0 is required for all Baccalaureate degrees.
- AAE requires a minimum of a 2.0 for major GPA. All AAE Required Engineering Major courses plus AAE Specialization courses and Selective courses count towards the AAE major GPA calculation.

Grade Requirements

A grade of C- or better is required in AAE 20300, AAE 20400, AAE 33800 (if taken), and all MA (Math) courses.

Pass/No Pass Policy

- General Education and Technical Selective courses may be taken as Pass/No Pass. All other courses within the AAE degree requirements are required to be taken for a grade.
- Study abroad courses may be approved for the Pass/No Pass grade mode. Consult with your academic advisor when planning your study abroad.

Transfer Credit Policy

If you are interested in registering for a course offered by a different institution, please review the information on the Purdue Office of the Registrar Transfer Credit website.

Courses listed as "#XXXX" are considered undistributed credit, or courses which do not have a Purdue equivalent. These courses cannot be used to meet AAE degree requirements.

*If you are an admitted transfer student, please review the above Transfer Credit website and then work with your academic advisor to determine exactly how your previous courses might apply.

Course Requirements and Notes

Students may double count in the following areas:

- UCC: Humanities for General Education

- UCC: Behavioral/Social Science for General Education
- UCC: Science, Tech, and Society for either Technical Selective or General Education
- Civics Literacy courses for General Education
- Minor and certificate courses for Technical Selectives, General Education, AAE Statistics Rule, math requirements, or AAE Specialization/Selectives

Students may not double count in the following areas

- Technical Selectives for AAE Specialization/Selectives

Students are allowed to repeat courses, regardless of the grade, up to 3 attempts per University regulations.

Notes

- Students in Aeronautical and Astronautical Engineering, BS cannot also enroll in the Mechanical Engineering, BS.

University Requirements

University Core Requirements

FOR A COMPLETE LISTING OF UNIVERSITY CORE COURSE SELECTIVES, VISIT THE UNIVERSITY SENATE WEBSITE.

- AI Working Competency (UCC: AI)
- Human Cultures: Behavioral/Social Science (UCC: BSS)
- Human Cultures: Humanities (UCC: HUM)
- Information Literacy (UCC: IL)
- Oral Communication (UCC: OC)
- Quantitative Reasoning (UCC: QR)
- Science #1 (UCC: SCI)
- Science #2 (UCC: SCI)
- Science, Technology, and Society (UCC: STS)
- Written Communication (UCC: WC)

Civics Literacy Proficiency Requirement

The Civics Literacy initiative is designed to develop civic knowledge of Purdue students to graduate a more informed citizenry. For more information visit the Civics Literacy - Office of the Vice Provost for Teaching and Learning - Purdue University.

To obtain the Civics Literacy credential, all Baccalaureate degree seeking students will complete the Civics Literacy Podcast pathway - Complete 12 podcasts created by the Purdue Center for C-SPAN Scholarship & Engagement that use C-SPAN material and pass the required quizzes per episode. After the pathway has been successfully completed, students will take the Civics Literacy knowledge test. Civics Literacy does not impact a student's GPA or bear any financial costs.

More details about each pathway and how to complete the requirement can be found on the Civics Literacy Student Dashboard in myPurdue.

Upper Level Requirement

- Resident study at Purdue University for at least two semesters and the enrollment in and completion of at least 32 semester hours of coursework required and approved for the completion of the degree. These courses are expected to be at least junior-level (30000+) courses.
- Students should be able to fulfill *most*, *if not all*, of these credits within their major requirements; there should be a clear pathway for students to complete any credits not completed within their major.

Sample Plan of Study

The following sample plan of study is a suggested arrangement of courses. Consultation with an advisor may result in an altered plan customized for an individual student.

Fall 1st Year

- Intro to Engineering I – Credit Hours: 2.00-4.00
- Written Communication (UCC: WC) or Oral Communication (UCC:OC) - Credit Hours: 3.00-4.00
- CHM 11510 - General Chemistry I **Credit Hours: 3.00**

- CHM 11520 - General Chemistry I - Laboratory (UCC: SCI) **Credit Hours: 1.00** or
- CHM 11530 - General Chemistry I - Virtual Laboratory (UCC: SCI) **Credit Hours: 1.00**

- MA 16100 - Plane Analytic Geometry And Calculus I (UCC: QR) **Credit Hours: 5.00** or
- MA 16500 - Analytic Geometry And Calculus I (UCC: QR) **Credit Hours: 4.00**

13-17 Credits

Spring 1st Year

- Intro to Engineering II - Credit Hours: 2.00-4.00
- Written Communication (UCC: WC) or Oral Communication (UCC:OC) - Credit Hours: 3.00-4.00
- Physics I - Credit Hours: 4.00
- CS 15900 - C Programming **Credit Hours: 3.00**

- MA 16200 - Plane Analytic Geometry And Calculus II (UCC: QR) **Credit Hours: 5.00** or
- MA 16600 - Analytic Geometry And Calculus II (UCC: QR) **Credit Hours: 4.00**

16-20 Credits

Fall 2nd Year

- AAE 20000 - Undergraduate Sophomore Seminar **Credit Hours: 0.00**
- AAE 20300 - Aeromechanics I **Credit Hours: 3.00** ♦
- MA 26100 - Multivariate Calculus (UCC: QR) **Credit Hours: 4.00** ♦
- MA 26500 - Linear Algebra (UCC: QR) **Credit Hours: 3.00** ♦
- MFET 16300 - Foundations In Geometric Modeling And Data Management **Credit Hours: 2.00**
- General Education I (UCC: HUM) - Credit Hours: 3.00-4.00

15 Credits

Spring 2nd Year

- AAE 20400 - Aeromechanics II **Credit Hours: 3.00** ♦
- AAE 20401 - Aeromechanics II Laboratory **Credit Hours: 1.00**
- AAE 25100 - Introduction To Aerospace Design **Credit Hours: 3.00** ♦
- MA 26600 - Ordinary Differential Equations (UCC: QR) **Credit Hours: 3.00** ♦

- ME 20000 - Thermodynamics I **Credit Hours: 3.00** ♦
- General Education II (UCC: BSS) - Credit Hours: 3.00

- PHYS 24100 - Electricity And Optics (UCC: SCI) **Credit Hours: 3.00** or
- PHYS 27200 - Electric And Magnetic Interactions (UCC: SCI) **Credit Hours: 4.00**

19-20 Credits

Fall 3rd Year

- AAE 30000 - Undergraduate Junior Seminar **Credit Hours: 0.00**
- AAE 30100 - Signal Analysis For Aerospace Engineering **Credit Hours: 3.00**
- AAE 33300 - Fluid Mechanics **Credit Hours: 3.00**
- AAE 33301 - Fluid Mechanics Laboratory **Credit Hours: 1.00**
- AAE 35200 - Structural Analysis I **Credit Hours: 3.00**
- MA 30300 - Differential Equations And Partial Differential Equations For Engineering And The Sciences **Credit Hours: 3.00** ♦
- General Education III - Credit Hours: 3.00

16 Credits

Spring 3rd Year

- AAE 33400 - Aerodynamics **Credit Hours: 3.00**
- AAE 34000 - Dynamics And Vibrations **Credit Hours: 3.00**
- AAE 36400 - Control System Analysis **Credit Hours: 3.00**

- AAE 33401 - Aerodynamics Laboratory **Credit Hours: 1.00** or
- AAE 35201 - Structural Analysis I Laboratory **Credit Hours: 1.00**

- AAE 33800 - Thermal Sciences **Credit Hours: 3.00** or
- AAE 33900 - Aerospace Propulsion **Credit Hours: 3.00**

- General Education IV - Credit Hours: 3.00

16 Credits

Fall 4th Year

- AAE 36401 - Control Systems Laboratory **Credit Hours: 1.00**
- AAE 40000 - Undergraduate Senior Seminar **Credit Hours: 1.00**
- AAE Engineering Specialization - Credit Hours: 3.00
- AAE Selectives - Credit Hours: 3.00
- AAE Statistics Selective - Credit Hours: 3.00
- AAE Technical Selective - Credit Hours: 3.00
- General Education V - Credit Hours: 3.00

17 Credits

Spring 4th Year

- AAE 42100 - Flight Dynamics And Control **Credit Hours: 3.00** or
- AAE 44000 - Spacecraft Attitude Dynamics **Credit Hours: 3.00**

- AAE 45000 - Spacecraft Design **Credit Hours: 3.00** or
- AAE 45100 - Aircraft Design **Credit Hours: 3.00**

- AAE Engr Specialization Selectives - Credit Hours: 6.00
- AAE Selectives - Credit Hours: 3.00
- General Education VI - Credit Hours: 3.00

18 Credits

Pre-Requisite Information

For pre-requisite information, log in to mypurdue.purdue.edu and click here.

World Language Courses

World Language proficiency requirements vary by program. The following list is inclusive of all world languages PWL offers for credit; for acceptable languages and proficiency levels, see your advisor. (ASL-American Sign Language; ARAB-Arabic; CHNS-Chinese; FR-French; GER-German;

GREK-Greek(Ancient); HEBR-Hebrew(Biblical); HEBR-Hebrew(Modern); ITAL-Italian; JPNS-Japanese; KOR-Korean; LATN-Latin; PTGS-Portuguese; RUSS-Russian; SPAN-Spanish)

Critical Course

The ♦ course is considered critical.

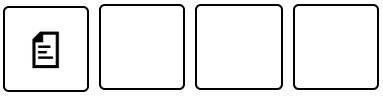
In alignment with the Degree Map Guidance for Indiana's Public Colleges and Universities, published by the Commission for Higher Education (pursuant to HEA 1348-2013), a Critical Course is identified as "one that a student must be able to pass to persist and succeed in a particular major. Students who want to be nurses, for example, should know that they are expected to be proficient in courses like biology in order to be successful. These would be identified by the institutions for each degree program."

Disclaimer

The student is ultimately responsible for knowing and completing all degree requirements. Consultation with an advisor may result in an altered plan customized for an individual student. The myPurduePlan powered by DegreeWorks is the knowledge source for specific requirements and completion.

Comparative information about Purdue University and other U.S. educational institutions is also available through the College Navigator tool, provided by the National Center for Education Statistics, and through the U.S. Department of Education College Scorecard.

Return to: Catalog Search



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